

Embodied Manipulation for Robotic Arms: Embodiment, Data, Learning, and Evaluation

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Robotic manipulation has advanced rapidly with the emergence of deep learning, foundation models, and large language models, giving rise to a diverse landscape of learning paradigms and system designs. Existing surveys, however, are predominantly organized around model families or methodological categories, which tend to obscure the embodied constraints that cut across these paradigms and shape manipulation capability in a stage-wise manner. This survey revisits robotic arm manipulation from an embodied perspective. We introduce an organizational framework centered on a chain of embodiment constraints, through which manipulation capability is understood as a capability formation process spanning data support, knowledge support, cognitive planning, policy learning, integration, evaluation, and deployment. Using this framework, we reorganize the literature beyond model-family labels and show how many seemingly paradigm-level differences are better understood as local responses to shared embodied constraints. Based on this view, we further analyze how limitations originating in embodied foundations accumulate in data support, become amplified in architectural design, complicate evaluation as a source of reliable upstream feedback, and open onto broader concerns of safety and trustworthiness in deployed embodied systems. By tracing these coupled pressures across stages, this survey aims to provide a more structured understanding of robotic manipulation and to clarify the central challenges for building scalable, deployable, and trustworthy embodied systems.

CCS Concepts: • **Computer systems organization** → **Robotic autonomy**; *Robotic control*; External interfaces for robotics.

Additional Key Words and Phrases: Embodied intelligence, robotic manipulation, perception, control, learning

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1 Introduction

Embodied intelligence starts from the premise that cognition is shaped by the dynamic coupling of brain, body, and environment [32]. More specifically, intelligent behavior must remain grounded in the continuous coupling of perception, action, and physical interaction in the real world [16]. Robotic arm manipulation provides an ideal platform for embodied intelligence, as it concentrates

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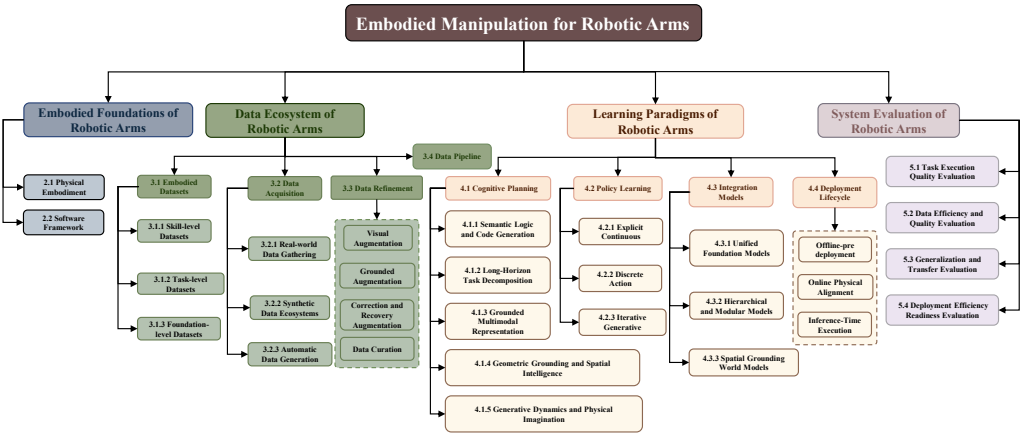


Fig. 1. Overview of manipulation intelligence for robotic arms, structured around system architecture, data ecosystem, architectural paradigms, and evaluation metrics, tracing connections from embodiment and data pipelines to cognitive planning, policy learning, integration models, deployment, and evaluation.

the constraints of perception, action, contact dynamics, and high-DoF control within a single physical system [85]. These characteristics have made robotic arms a central platform for embodied intelligence research, and successive advances in deep learning and, more recently, foundation models have progressively reshaped the methodological landscape of robotic manipulation. Several recent surveys cover adjacent parts of this landscape, including unified abstractions for robotic manipulation [9], review of vision-language-action (VLA) models for real-world robotics [83], object-centric embodied learning [221], and broader embodied AI systems built on LLMs and world models [109]. However, existing surveys on robotic manipulation are predominantly structured around model paradigms. Without a structure more explicitly grounded in embodiment, the resulting taxonomies often become difficult to interpret and more complex than necessary. Under such conditions, local methodological variants are easily treated as separate categories rather than as design choices within a shared structure.

This survey is organized around one central idea: robotic arm manipulation should be understood as a process of capability formation under embodiment constraints. From this perspective, the survey makes three closely connected contributions. First, it provides an explanatory overview of robotic arm manipulation as a process in which capability is progressively formed, supported, and limited under embodiment constraints. This perspective shifts attention from isolated algorithmic categories to the cumulative role of physical, perceptual, and control constraints in shaping manipulation performance. Second, it develops a layered taxonomy that organizes the literature around data, learning, and evaluation as interdependent stages under embodiment constraints. This situates local design variations within a shared analytical structure, enabling meaningful comparison across approaches. Third, it uses the resulting organization to reveal a challenge structure that accumulates across the full system stack of robotic manipulation for robotic arms.

Structure of the Paper: Section 2 reviews physical embodiment and software infrastructure. Section 3 examines the data ecosystem from datasets through acquisition, synthesis, refinement, and pipelines. Section 4 analyzes cognitive planning, policy learning, integration models, and deployment. Section 5 synthesizes evaluation methodologies, and Section 6 concludes with open challenges and future directions. Figure 1 summarizes this organization.

2 Embodiment Foundations of Robotic Arms

For robotic arm platforms, the foundations of manipulation lie in two tightly connected aspects of embodiment. Physical embodiment defines how interaction is physically structured through contact, sensing, action, and platform constraints. Software framework makes these embodied capabilities operational by supporting data capture, communication, low-level control, and the formalization of robotic experience for downstream learning. The rest of this section is organized accordingly. Section 2.1 and Section 2.2 examine these two aspects in turn.

2.1 Physical Embodiment

Physical embodiment shapes robotic manipulation before learning starts. In robotic arm systems, it governs how contact can be made, and how these conditions can be converted into controllable behavior. We begin with the manipulation interface, through end effectors and sensors, then turn to robotic arms as the platforms that organize multi-DoF motion into executable manipulation, and finally discuss how arms are integrated into larger mobile and whole-body systems. Figure 2 summarizes this taxonomy. From this perspective, physical embodiment can be seen as the source of the constraints that define manipulation objectives, data requirements, and methodological choices.

2.1.1 End Effectors and Sensors. End effectors shape the contact space of manipulation by determining how contact is formed, maintained, and transformed during interaction. Different end-effector designs therefore differ not only in capability, but in the contact regime through which manipulation is organized. Jaw grippers, such as the OnRobot RG2/RG6 series, favor stable and mechanically constrained contact, making grasping, alignment, and release easier to control and predict. Dexterous hands, such as the SCHUNK SVH and Tesollo DG-5F-M, allow contacts to change more freely during manipulation and support richer in-hand interaction, but they also make behavior harder to reduce to a small set of repeatable contact patterns. Soft hands, such as SpiRobs [175], address uncertainty differently by using compliance to absorb part of the interaction variability in the hardware itself.

Sensors shape the observation space of manipulation by providing access to task-relevant information during execution. The central task of sensors is to ensure the information required at different stages of interaction is observable. Before contact, the dominant uncertainties concern object geometry, pose, and scene layout, so visual sensing is often sufficient for coarse localization, alignment, and approach. During contact, however, task-relevant information becomes increasingly local and interaction-dependent, including slip, deformation, insertion error, and force misalignment. These states are often critical to task success, yet they are only weakly reflected in external visual observations. For this reason, tactile, force, and proprioceptive sensing become critical once task-relevant information is expressed primarily through contact itself. Sensors such as GelSight [98], DIGIT [89], and IMU make different portions of this observation space accessible by exposing local contact geometry, contact transients, and motion signals that are difficult to recover from external vision alone.

End effectors and sensors shape how a robotic arm interacts with its environment. Through contact space and observation space, they bound the otherwise open-ended complexity of manipulation into structured constraints, making the discussion that follows more tractable.

2.1.2 Fixed-base Robotic Arms. In common robotic manipulation settings, fixed-base robotic arms provide controllable multi-DoF motion and serve as the platform through which end effectors and sensors are brought into execution. They organize motor commands into structured sequences of actions. Through active motion and contact, these action sequences induce changes in task state

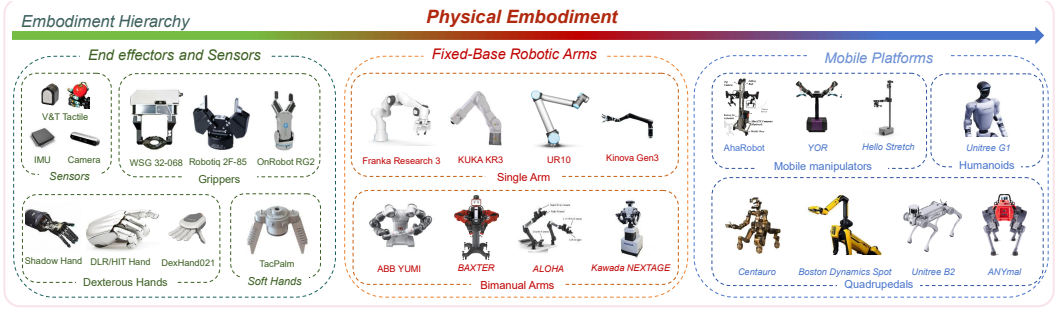


Fig. 2. Overview of physical embodiments for robotic manipulation. The figure covers end effectors, sensors, stationary platforms, and mobile platforms.

and observation during execution. Hence, we will focus on how robotic arms transform motor commands into reliable action sequences.

In single-arm settings, such as KUKA LBR iiwa [123], the action structure is relatively localized. It is typically organized around a compact set of objectives through motor commands, including establishing a stable grasp, aligning the object to a desired pose, and completing a local contact transition. Since their execution structure is relatively localized and compact, motion sequences and state transitions in single-arm settings are relatively easier to collect, model, and evaluate. Consequently, single-arm systems remain the dominant setting for teleoperation, policy benchmarking, and manipulation with high precision.

In bimanual settings, such as PR2 [181], manipulation requires coordination across multiple arms. Once multiple manipulators act on the same object, the task becomes coordinated transformation under shared kinematic and force constraints. This coordination enables the system to perform tasks that require coordinated force regulation or the handling of large, heavy, or deformable objects. As a result, bimanual systems are increasingly used in assembly, tool use, and human-robot collaboration.

These manipulation settings arise from practical task demands, and in turn become part of what shapes and constrains the ecosystem of embodied intelligence. Action sequences, together with the state transitions they induce, will form an important part of data support discussed in Section 3.

2.1.3 Mobile Platforms. Mounting robotic arms on mobile platforms expands the scope of manipulation. In wheeled mobile manipulators such as Mobile ALOHA [43], Stretch [84], arm action needs to coordinate with base motion and navigation constraints. In legged platforms and humanoids, manipulation must be executed together with balance, posture regulation, and system-level support conditions. In conclusion, robotic arms, as a central component of robotic manipulation, are increasingly being integrated into larger support systems, thereby exhibiting greater capability and broader potential.

2.2 Software Framework

The software framework of robotic manipulation provides the infrastructure that makes embodied capabilities operational in practice. During runtime, it supports signal capture and transport, coordinates communication across distributed components, and sustains low-level control. Beyond execution, it also provides the data interfaces through which robotic experience is formalized for data and downstream learning.

Table 1. Three-tier software architecture for embodied manipulation intelligence for robotic arms. The software stack is organized into three layers. L0 is the Capture Layer for hard real-time motor control with microsecond-level determinism. L1 is the Transfer Layer for high-throughput, low-latency inter-process and inter-device communication. L2 is the Orchestration Layer for distributed orchestration across robots, edge devices, cloud resources, and fleets. The Profile column reports interrupt or IPC latency for L0 and L1 systems, and deployment or orchestration characteristics for L2 systems.

Framework	Type	Core Mechanism	Key Contribution	Profile	Scope
L0: Capture Layer					
VxWorks [192]	Monolithic	Priority Preemptive	Deterministic RTOS kernel with priority-based preemptive scheduling, designed for low-latency and low-jitter execution in control systems.	<10 μ s	Industrial
QNX [59]	Microkernel	Spatial Isolation	Microkernel RTOS that isolates drivers and system services, improving fault containment while preserving deterministic scheduling.	<5 μ s	Safety-critical
L1: Transfer Layer					
ROS 2 [116]	Decentralized	DDS	Middleware built on DDS, providing configurable QoS policies for reliability, deadline, and liveness in distributed robotic systems.	1–10 ms	General Robotics
Dora-rs [214]	Dataflow	Apache Arrow	Rust-based dataflow runtime using Apache Arrow and shared memory for low-latency, zero-copy exchange across modular robotic processes.	<50 μ s	Low-latency AI
L2: Orchestration Layer					
Zenoh-Flow [11]	DAG	Zenoh Protocol	Communication and query protocol that unifies pub/sub, storage, and distributed computation across devices ranging from embedded nodes to edge and cloud systems.	Cloud	Distributed
FogROS2 [70]	Cloud	ROS 2 Graph	ROS 2 extension which enables selected components of a robotic graph to be offloaded while retaining local reactive control.	Hybrid	Robot-as-a-Service

2.2.1 Modern Robotics Middleware and Operating Systems. The software stack of embodied manipulation systems exists to make embodiment data accessible: ensuring signals are captured with sufficient temporal fidelity, transported without bottleneck, and organized at scale across heterogeneous hardware and compute. We organize this stack into three tiers, as summarized in Table 1. The Capture Layer (L0) ensures the temporal determinism required for reliable signal acquisition from sensors and actuators. Industrial stacks rely on real-time operating system (RTOS) kernels such as VxWorks [192] and QNX [59] for bounded interrupt latency and fault isolation, while Zephyr extends the same guarantees to edge devices. The Transfer Layer (L1) moves captured data between processes, devices, and model components. ROS 2 provides the baseline connectivity through decentralized Data Distribution Service (DDS) [116], while the growing cost of moving multimodal embodiment data has driven zero-copy designs such as Dora-rs [214]. The Orchestration Layer (L2) organizes data flows across robots, edge devices, and cloud services at the system level. Zenoh-Flow and FogROS2 are designed with deployment, routing, and compute offloading as core system concerns [11, 70], while RoboOS couples multimodal planning to reusable skill libraries across heterogeneous embodiments [159].

2.2.2 Low-level Motion Planning and Control. While middleware provides the infrastructure for data capture and transport, low-level motion planning and control determine how physical embodiment is actuated during execution. For end effectors, impedance control enables compliant interaction by regulating the relationship between positional deviation and contact force, and it is the standard approach for assembly, tool use, and other forms of contact-rich manipulation. For fixed-base

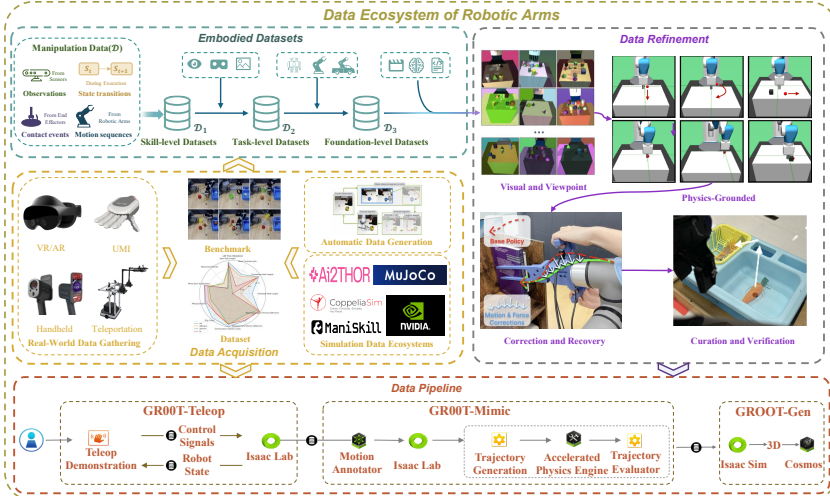


Fig. 3. Overview of the embodied data ecosystem, organized into embodied datasets, data acquisition, data refinement, and data pipelines. The taxonomy shows how data are collected, structured, refined, and incorporated into downstream learning and deployment pipelines.

arms, this layer resolves operational objectives into joint motions and torques through kinematic mapping, rigid-body dynamic compensation, and enforcement of joint and collision constraints. For mobile platforms and humanoids, where manipulation must be coordinated with locomotion, balance, and whole-body motion, optimization-based approaches such as MPC and WBC extend this control layer by solving tracking, feasibility, and safety objectives within a unified constrained optimization problem online at the control rate [142].

Beyond execution, software frameworks also define how robotic experience is recorded and structured for learning. Common formats range from runtime logs and telemetry records to trajectory dataset schemas such as RLDS and HDF5-based demonstrations, which organize observations, actions, rewards, and metadata for imitation learning and offline reinforcement learning.

3 Data Ecosystem of Robotic Arms

Having clarified how embodiment constrains robotic manipulation through contact space, observation space, and execution, we now turn to the data ecosystem through which these constraints become learnable. In robotic manipulation, we denote robotic manipulation data by $\mathcal{D}$, which comprises four components generated as the robot interacts with the world: **motion sequences** produced by the robotic arms, **sensory observations** provided by sensors, **contact events** realized through end effectors, and the resulting **state transitions** induced during execution. Progress therefore depends not only on model design, but also on how $\mathcal{D}$ is organized, collected, refined, and scaled across tasks, scenes, and embodiments. This chapter begins with embodied datasets in Section 3.1 and then examines the broader data processes associated with them, including data acquisition in Section 3.2, data refinement in Section 3.3, and large-scale dataset pipelines in Section 3.4. Together, these elements provide a structured view of the data ecosystem for robotic arm manipulation, spanning embodied datasets, data acquisition, data refinement, and large-scale data pipelines, as summarized in Figure 3.

Table 2. **Chronological Overview of Skill-level Datasets.** Evolution reflects the shift from planar grasping rectangles to dense 6-DoF poses, dexterous hands, language-grounded reasoning, and diverse motor primitives (pushing, articulation, deformable manipulation, assembly).

Dataset	Year	Skill Type	Scene Complexity	Objects	Domain	Scale	Modality	Lang.
Early Geometric Grasping								
Cornell [93]	2011	Rect. Grasp (2D)	Single Object	240	Real	1k imgs	RGB-D	×
Dex-Net 2.0 [117]	2017	Parallel-Jaw Grasp	Single Object	1.5k	Sim	6.7M grasps	Depth	×
Dense 6-DoF Grasping								
GraspNet-1B [39]	2020	6-DoF Grasp	Heavy Clutter	88	Real	97k imgs (1.2B grasps)	RGB-D	×
GraspClutter6D [8]	2025	6-DoF Grasp	Cluttered	200	Real	52k imgs (9.3B grasps)	RGB-D	×
Dexterous Grasping								
DexGraspNet 2.0 [207]	2024	Dexterous Grasp	Multi-Object	5,355	Sim	427M grasps	Point Cloud	×
Dex1B [194]	2025	Dexterous Grasp	Single Object	1M	Sim	1B grasps	Point Cloud	×
Language-Grounded Grasping								
Grasp-Anything [165]	2023	Rect. Grasp + Push	Multi-Object	3M	Syn	1M imgs (600M grasps)	RGB	✓
ReasoningGrasp [80]	2024	6-DoF Grasp	Multi-Object	64	Real	99M grasps	RGB-D	✓
Diverse Motor Primitives								
Omnipush [12]	2019	Pushing (Planar)	Single Object	250	Real	62.5k pushes	RGB-D	×
PartNet-Mobility [183]	2020	Articulated (Open/Close)	Single Object	2,346 models	Sim	2,346 articulated assets	3D / PC	×
FurnitureBench [56]	2023	Assembly (Precise)	Tabletop	Furniture	Sim+Real	5.1k demos (220 hrs)	RGB	×

3.1 Embodied Datasets

Through a systematic review of the literature, we observe that embodied manipulation datasets have evolved toward progressively broader forms of support for learning: from preserving local interaction structure, to capturing how robotic manipulation unfolds over temporally extended execution, and ultimately to supporting transfer across heterogeneous embodiments. We abstract this progression into three increasingly broad tiers of embodied datasets: skill-level datasets ($\mathcal{D}_1$), which preserve local interaction structure; task-level datasets ($\mathcal{D}_2$), which preserve these interactions over temporally extended execution; and foundation-level datasets ($\mathcal{D}_3$), which preserve transfer across heterogeneous embodiments at scale. In the remainder of this section, we review embodied manipulation datasets according to these three levels, Section 3.1.1 for skill-level datasets, Section 3.1.2 for task-level datasets, and Section 3.1.3 for foundation-level datasets. As the scope of $\mathcal{D}$ expands across these tiers, the implied knowledge support $\mathcal{K}$ expands accordingly. We will discuss the role of $\mathcal{K}$, and how it supports different learning paradigms, in greater detail in Chapter 4.

3.1.1 Skill-level Datasets. The development of skill-level datasets reflects a steady expansion of $\mathcal{D}_1$, from planar grasp annotations, to dense 6-DoF grasp synthesis, to dexterous hand configurations and other interactions such as pushing, articulation, and assembly. Table 2 lists representative datasets along this tier. Early grasping benchmarks such as Cornell [93] labeled graspable regions as 2D oriented rectangles, covering vertical parallel-jaw approaches. Dex-Net 2.0 [117] generated 6.7 million synthetic grasp examples labeled with analytic grasp metrics, enabling large-scale grasp learning without collecting training data through physical trials. GraspNet-1B [39] annotated 1.2 billion 6-DoF grasps across cluttered real-world scenes, and GraspClutter6D [8] extended this to denser clutter configurations. For multi-fingered hands (>20 DoF), DexGraspNet 2.0 [207] synthesized 427M grasps via diffusion-based generation (90.7% real-world success), and Dex1B [194] reached one billion grasps. Tactile modalities complement vision: T-DEX [49] showed that 2.5 hours of self-supervised tactile play data improves dexterous task success by 1.7× over baselines with only vision, capturing slip and reorientation signals invisible to cameras. Beyond grasping, Omnipush [12] records planar pushing dynamics across 250 objects, PartNet-Mobility [183] provides part interaction data for articulated objects, and FurnitureBench [56] supplies 5,000 teleoperated assembly trajectories with a paired simulator.

Table 3. **Representative task-level embodied datasets collected from real-world physical trajectories.**

Traj. denotes the number of recorded demonstrations or episodes; Views denotes the number of camera views available; Tasks denotes the number of distinct task types; Scenes denotes the number of distinct environments or scene settings; Lang. indicates whether language annotations or task instructions are provided; Depth indicates whether depth observations are provided; Calib. indicates whether camera calibration or equivalent calibration metadata is available; Form. indicates the primary release format, such as raw files, HDF5, TFDS, or RLDS; Platform denotes the robot platform used for data collection; Collection Mode summarizes the acquisition protocol; and Format denotes the primary release format. In the table, Teleop. = teleoperation, and H/S Hybrid = a combination of human demonstrations and scripted data collection.

Dataset	Year	Traj.	V	Tasks	Scenes	Lang.	Depth	Calib.	Platform	Collection	Form.
MIME [148]	2018	8.3k	3	20	1	×	✓	×	Baxter	Teleop.	Raw
RoboTurk [119]	2018	2.1k	2	3	1	×	✓	×	Panda	Teleop.	HDF5
MT-Opt [81]	2021	800k	1	12	1	×	×	×	Google Arm	Mixed	TFDS
BC-Z [72]	2021	26k	1	100	1	✓	×	×	EDR	Teleop.	TFDS
RT-1 Dataset [15]	2022	130k	1	700+	2	✓	×	×	EDR	Teleop.	TFDS
Dobb-E [146]	2023	1.5k	1	109	22	×	✓	×	Stretch	iPhone	TFDS
BridgeData V2 [166]	2024	60k	4	13	24	✓	✓	×	WidowX	H/S Hybrid	Raw
DROID [86]	2024	76k	3	84	564	✓	✓	✓	Panda	Teleop.	RLDS
FMB [114]	2025	22k	4	2	1	✓	✓	✓	Panda	Mixed	Raw

3.1.2 Task-level Datasets. Task-level datasets evolved $\mathcal{D}_2$ by extending isolated motor primitives into trajectories aligned with language instructions, gradually increasing semantic coverage, environmental diversity, and calibration quality across collection sites. Table 3 lists representative datasets along this tier. MT-Opt [81] collected 800k trajectories through autonomous exploration but used task IDs to index actions, without language supervision. BC-Z [72] introduced language-conditioned supervision by pairing 26k trajectories with 100 semantic verbs, and RT-1 Dataset [15] extended this to 130k trajectories spanning 700+ instructions. However, despite their growing scale, these early datasets were collected in a limited range of settings within one or a few research institutions, which restricted environmental diversity. BridgeData V2 [166] widened scene coverage to 24 environments via collection in several locations, DROID [86] coordinated 18 institutions across 564 scenes with calibrated extrinsics, and Dobb-E [146] accessed 22 domestic environments using only an iPhone mounted on a Stretch robot. FMB [114] added force-torque calibration across 22k contact-rich assembly trajectories.

3.1.3 Foundation-level Datasets. Foundation-level datasets expand $\mathcal{D}_3$ by organizing robot experience across platforms into a unified layer of support for cross-embodiment learning, with increasing emphasis on alignment, standardization, and multimodal consistency. Table 4 summarizes representative datasets in this tier. RoboNet [35] established an early multi-robot dataset with 15 million video frames collected across seven robot platforms. OXE [131] scaled this direction by unifying over one million trajectories from 22 robot platforms through the RLDS protocol. As this tier scales, standardization and multimodal alignment become equally important. ARIO [176] provides a unified large-scale dataset with approximately three million episodes and consistent timestamp alignment across heterogeneous sources, while RH-20T [40] contributes over 110k contact-rich manipulation sequences with synchronized visual, force, audio, and action signals. RoboMIND [179] extends this tier toward standardized multi-robot task datasets, pairing 107k trajectories across four robot types with failure annotations, and RoboMIND 2.0 [62] scales this to over 310k trajectories across six robot platforms and 739 tasks. AgiBot World Beta [17] pushes this tier toward humanoid and dual-arm manipulation, providing over one million trajectories from 100 robots across 100+ real-world scenarios. As robot data demand outpaces collection speed, some pipelines supplement trajectory datasets with egocentric human video such as Ego-Exo4D [47].

Table 4. **Foundation-level Datasets for Multi-Robot Manipulation.** # Traj. = trajectory count; # Embod. = distinct robot platforms/configurations when explicitly reported; Embod. Types: S = single-arm, B = bimanual, M = mobile manipulator, H = humanoid, D = depth, F = force/torque, A = audio; Lang. = language supervision; Action: EE = end-effector, Emb. = embodiment-specific, WB = whole-body; Alignment: how multi-robot differences are reconciled.

Dataset	Year	# Traj.	# Embo.	Types	Source	Modalities	Lang.	Action	Alignment
RoboNet [35]	2019	162k	7	S	Scripted	RGB	×	EE	Interface-level
RH-20T [40]	2023	110k	7	S	Teleop	RGB, D, F, A	✓	Emb.	Calibration-level
OXE [131]	2023	1M+	22	S, B, M	Aggregated	RGB	✓	EE	Schema-level
ARIO [176]	2024	3M	N/A	S, B, M	Aggregated	RGB, D	✓	Emb.	Temporal
AgiBot World [17]	2025	1M+	100+	B, H	Teleop	RGB, D, F	✓	WB	Pipeline-level

3.2 Data Acquisition

The previous section distinguished dataset tiers by the support they provide. However, acquiring such support is not straightforward. Unlike generic datasets, embodied manipulation data must record action sequences under the constraints of observation space and contact space. Collecting data that is both physically grounded and rich enough to support learning at scale is difficult. This section therefore examines three main routes by which embodied interaction is converted into $\mathcal{D}$ through real-world collection in Section 3.2.1, simulation ecosystems in Section 3.2.2, and automatic data generation in Section 3.2.3.

3.2.1 Real-world Data Gathering. Since real-world collection is conducted under concrete hardware and scene constraints, its efficiency and coverage are constrained by the robotic arm platform, end effectors, and execution environments. Recent research can be read as a gradual effort to weaken this coupling while retaining useful observation, contact, and execution structure. Direct teleoperation on the deployment robot provides a straightforward way to acquire such data, and it forms the basis of many early approaches. RoboTurk [119] built a teleoperation interface that enabled demonstrations to be collected from geographically distributed users over the internet. ALOHA [43, 217] introduced a low-cost teleoperation setup for bimanual and later mobile whole-body manipulation, making long-horizon and contact-rich data collection more accessible. Teleoperation preserves the coupling between observation space, contact space, and arm-centric execution most faithfully, but it is also tightly constrained by hardware access and human labor. For this reason, later work moves toward more portable and reusable data-collection interfaces, so that demonstrations can be transferred more easily across robot embodiments and deployment setups. UMI [29] used handheld grippers and consumer cameras to capture demonstrations in everyday environments and then retargeted them offline to the target robot platform, while DexCap [168] pushed this approach toward dexterous manipulation by recording high-dimensional human hand motion for transfer to multi-fingered robotic manipulators. Even so, these interfaces are limited to narrower robotic arms and still rely on teleoperation. This motivates a further step toward egocentric and video-based supervision by retargeting human demonstrations back into robotic arm execution. EgoMimic [82] used egocentric human video to enable large-scale data capture without requiring robot hardware during collection, and AV-ALOHA [31] showed that such data become more useful when the demonstrator’s viewpoint is perceptually aligned with the robot’s onboard observation space.

3.2.2 Simulation Data Ecosystems. Compared with real-world collection, simulation weakens the dependence of data acquisition on specific robotic arm platforms, end-effector and sensor setups, and scene conditions. This makes it easier to define different formulations of $\mathcal{D}$ for robotic arm manipulation under more explicit assumptions. In this section, MuJoCo, SAPIEN, and NVIDIA Isaac are treated as three representative directions because they align naturally with different tiers of

Table 5. **Simulation infrastructures for embodied manipulation.** Platforms are grouped into three main ecosystems (MuJoCo, SAPIEN, Isaac) and specialized platforms for specific task categories or physical regimes.

Simulator	Features	Capabilities & Modalities	Key Benchmarks & Datasets
I. The Core Simulation Ecosystems			
MuJoCo	Stable analytic contact solver; MJX hardware acceleration; Differentiable dynamics.	Physical Aligned Data: Trajectories for fine-grained motor control. Modalities: Proprioception, Contact forces, Kinematics, RGB.	Yu et al. [197], LIBERO [106], [126]
Isaac Suite	PhysX5; RTX rendering; Generative synthesis.	Generative Scalable Data: Large-scale synthetic and high-DoF control. Modalities: RGB-D, LiDAR, Segmentation, Bounding boxes, Force/Torque, Event camera.	Isaac Lab [122], Behavior-1k [94], Arnold [45]
SAPIEN	High-fidelity articulated dynamics; Part mobility; Vulkan rendering.	Semantic Interaction Data: Interactions across articulated categories and reasoning over articulated parts. Modalities: RGB-D, Masks, Contact forces, Articulated object states.	ManiSkill3 [160], SIMPLER [100], RoboTwin [125]
II. Specialized Task Platforms			
PyBullet / Coppeliasim	Lightweight legacy physics; Stable IK solvers; Hardware and ROS abstraction.	Algorithmic Baseline Data: Logic planning and action primitives. Modalities: RGB-D, Joint states, Contact forces, Proximity sensors, Force/Torque.	CALVIN [121], RLBench [71]
Habitat / AI2-THOR	Procedural indoor scene generation; Scalable visual navigation environments.	Embodied Navigation Data: Semantic search and indoor reasoning. Modalities: RGB-D, Segmentations, Object states, Agent pose, Audio.	OpenEQA [118], ALFRED [152]

$\mathcal{D}$ for robotic arm manipulation, while other platforms serve for specialized applications. Table 5 summarizes these design tendencies. MuJoCo [163] is well aligned with $\mathcal{D}_1$ data, where stable local interaction and arm-centric rollouts matter most for robotic arm execution. Meta-World [197] introduced a widely used simulated benchmark for multi-task and meta-reinforcement learning across 50 robotic manipulation tasks. Later benchmarks such as LIBERO [106] and RoboCasa [126] added lifelong-learning curricula and richer household scenes. MJX and MuJoCo Playground support large-batch rollouts, differentiable simulation, and sim-to-real analysis within the same stack [201]. SAPIEN [183] is particularly suitable for $\mathcal{D}_2$ support, as articulated object states and part-level annotations support structured, temporally extended interaction modeling for robot manipulation. PartNet-Mobility provides articulated assets with annotated joints, ManiSkill builds benchmark suites on this basis [160], and SIMPLER uses the same infrastructure for sim-to-real evaluation [100]. NVIDIA Isaac is the platform best aligned with $\mathcal{D}_3$, particularly in settings that demand multimodality, scale, and broad platform coverage [122]. Representative embodied benchmarks built on NVIDIA Isaac Sim include BEHAVIOR-1K, which targets large-scale household activities, and ARNOLD, which focuses on language-grounded manipulation with continuous object states [45, 94]. Beyond these three ecosystems, several other platforms are still used for specialized applications of robotic arm manipulation. PyBullet and Coppeliasim support manipulation baselines such as RLBench and CALVIN [71, 121]. Habitat and AI2-THOR help compensate for the limited semantic support available in mainstream manipulation simulators, supporting benchmarks such as ALFRED and OpenEQA [118, 152]. SoftGym supports deformable object manipulation benchmarks that rigid-body simulators still capture only weakly [105]. By reducing dependence on physical robot hardware, simulation improves experimental efficiency and allows more effort to be directed toward algorithmic questions above the arm hardware layer. However, simulation cannot fully capture real-world physics and hardware-specific behavior, and the target arm itself is not involved

in the data collection loop. Consequently, the resulting $\mathcal{D}$ for robotic arm learning is often not fully faithful to robotic manipulation, leading to a substantial sim-to-real gap.

3.2.3 Automatic Data Generation. Building on the environments discussed above, Automatic data generation synthesizes different parts of $\mathcal{D}$, the structured record of embodied execution. Based on the content of this structured record, we discuss automatic data generation in three forms: scene generation synthesizes sensory observations, demonstration synthesis generates motion sequences together with contact events, and generative world models synthesize state transitions along with their future rollouts. Scene generation synthesizes sensory observations by widening the scene and semantic conditions under which they are produced within the observation space. ProcTHOR [36] generated large-scale procedural indoor environments, Holodeck [193] introduced semantic scene generation for embodied environments, and RoboGen [172] connected asset retrieval, scene construction, and policy training within one automated pipeline. Demonstration synthesis generates motion sequences together with contact events by generating additional robotic arm trajectories from limited source demonstrations or task specifications. MimicGen [120] showed that one demonstration can be expanded into many valid trajectories through spatial transformation, and GenSim [170] moved this idea to task generation by using LLMs to produce simulation code and reward structure. Generative world models synthesize state transitions along with their future rollouts by learning predictive models that can synthesize additional transitions from existing data. Cosmos [128] extends this paradigm to web-scale video data, training video foundation models to acquire physical priors from large-scale visual experience. IRASim [227] brings this into the real-robot setting through video world modeling of fine-grained robotic manipulation. DreamGen uses learned world models as both data generators and training environments [73]. Automatic data generation, however, synthesizes $\mathcal{D}$ by recombining or rolling forward an existing environment. The errors and limitations of that environment are also amplified.

3.3 Data Refinement

In real-world settings, available $\mathcal{D}$ is often still limited in scale and uneven in quality, which motivates further data refinement and augmentation. We organize these methods around four mechanisms: visual and viewpoint augmentation in Section 3.3.1 broadens sensory observations, physically grounded augmentation in Section 3.3.2 preserves the executability of motion sequences together with contact events under kinematic and contact constraints, correction and recovery augmentation in Section 3.3.3 supplements state transitions with the failure-recovery data, and data curation and verification in Section 3.3.4 determine whether existing $\mathcal{D}$ is still usable for learning and how low-quality data can be filtered, reweighted, or curated.

3.3.1 Visual and Viewpoint Augmentation. Visual and viewpoint augmentation expands existing demonstrations along two main axes: semantic appearance variation and camera transformation, both without requiring new robot rollouts. Early work such as GenAug and ROSIE used generative image models to vary textures, objects, and scene composition in order to widen the visual distribution seen during imitation learning [25, 198]. Later methods moved beyond appearance randomization to augmentations that explicitly model changes in camera viewpoint: RoVi-Aug augments both robot appearance and camera pose while preserving action labels and contact consistency under novel views [24]. A complementary line extends supervision beyond robot demonstrations: VISTA [161] enables multi-view synthesis from a single view for policy learning, H2R [95] converts egocentric human video into robot-centric pretraining data by replacing human hands with rendered end effectors, and Pan et al. [132] combine novel view synthesis with trajectory warping to derive diverse training trajectories from a single demonstration.

3.3.2 Physically Grounded Augmentation. Augmenting physical trajectories (e.g., shifting a grasp 5 cm laterally) requires that the result still obey kinematic limits, collision constraints, and contact dynamics. CP-Gen [104] integrates kinematic boundaries with contact dynamics for one-shot trajectory augmentation. Yang et al. [191] combine VR human demonstrations, retargeting, and trajectory optimization to maintain dynamic feasibility across embodiments. CyberDemo [169] transfers simulated human demonstrations to real-world dexterous policies via large-scale augmentation in simulation, using minimal real data. Beyond geometric feasibility, RoCoDA uses symmetry to distinguish perturbations that leave the policy invariant from those that induce equivariant action changes.

3.3.3 Correction and Recovery Augmentation. Most datasets lack failure-recovery examples, making policies fragile when they encounter OOD cases. DMD [212] uses diffusion models to synthesize corrective observations from perturbed states, enabling recovery without physical collection. Complementing these offline methods, IntervGen [61] and CR-Dagger [186] systematize online human intervention during autonomous rollouts, so that sparse human corrections can be used to train policies to recover from mistakes.

3.3.4 Data Curation and Verification. As robot datasets scale in size and heterogeneity, the bottleneck shifts from data quantity to data utility. Existing methods differ primarily in how they estimate data value. ReMix [54] learns how to weight different dataset domains during pretraining so as to improve robustness across downstream domains. SCIZOR [215] prunes transitions that are weakly informative for task progress or redundant with other samples. CUPID [2] uses influence estimates to assess the contribution of each demonstration to a policy’s closed-loop performance, and then uses these estimates for filtering and data selection.

3.4 Data Pipeline

In current practice, $\mathcal{D}$ is increasingly managed over a unified lifecycle, where collection, generation, augmentation, and refinement are treated as parts of one integrated production lifecycle. Existing systems reflect different ways of organizing this full lifecycle of $\mathcal{D}$. RoboTwin [125] keeps synthetic data tethered to measured physical setups by matching dual-arm kinematics and sensor noise, and its second version adds task synthesis for automatic scenario generation. RoboVerse [44] addresses fragmentation across simulators and datasets by providing a standardized MetaSim backbone spanning many tasks and assets. At larger scale, AgiBot World Colosseo [17] couples large-scale humanoid collection with augmentation in simulation so that real and synthetic streams remain mutually useful, while NVIDIA Isaac GR00T [129] packages synthetic data generation, gathering, refinement, and sim-to-real alignment into a humanoid stack built on Isaac infrastructure. Supporting these pipelines, AIRSPEED [182] decouples collection and generation from specific hardware backends, LeRobot [19] turns fragmented real-world trajectories into reusable repositories, and MimicLabs [141] studies how quantity, diversity, and quality trade off as pipelines scale. Taken together, these infrastructures move $\mathcal{D}$ production toward automated and reproducible workflows.

4 Learning Paradigms of Robotic Arms

Learning paradigms are centered on $\mathcal{D}$, introduced in Section 3.1, from which they derive the knowledge support $\mathcal{K}$ required for robotic manipulation under embodiment constraints. We distinguish two primary paradigms according to how they acquire and organize $\mathcal{K}$: cognitive planning, denoted by B , and policy learning, denoted by P . This distinction is reflected in the components of $\mathcal{D}$ that each paradigm primarily emphasizes. Cognitive planning (B) relies mainly on sensory observations and state transitions to construct intermediate artifacts and specify actions, whereas policy learning (P) relies more on contact events and motion sequences to produce executable motor

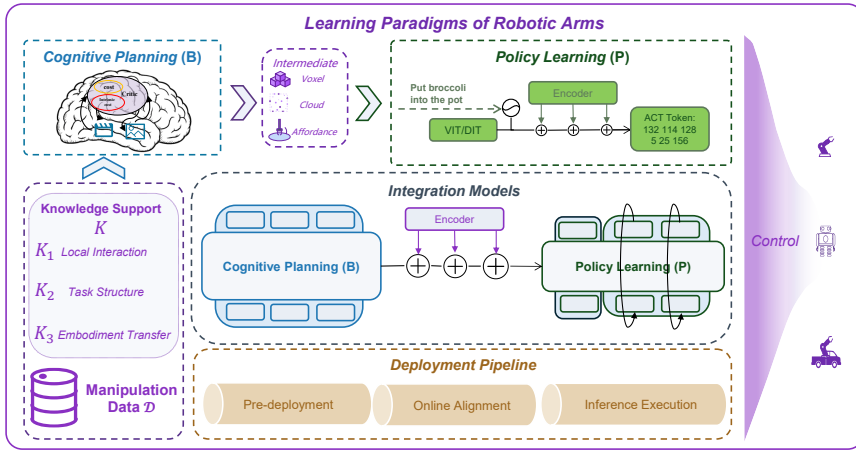


Fig. 4. Overview of learning paradigms for robotic arm manipulation. The figure organizes the pipeline into cognitive planning, policy learning, integration models, and deployment stages, linked by multimodal inputs and downstream control across robotic arm embodiments.

behavior under different action representations. Integration models couple these two pathways into unified architectures, either explicitly or implicitly. At the level of knowledge support, we further distinguish three scopes. $\mathcal{K}_1$ supports learning local interaction, $\mathcal{K}_2$ supports learning task structure for execution, and $\mathcal{K}_3$ supports learning transferable structure across embodiments. Section 4.1 introduces cognitive planning paradigms, Section 4.2 covers policy learning paradigms, Section 4.3 examines how the two pathways are coupled, and Section 4.4 discusses deployment through a three-stage pipeline. Figure 4 summarizes this learning ecosystem.

4.1 Cognitive Planning: The Embodied Brain

Cognitive planning places greater emphasis on sensory observations and state transitions in $\mathcal{D}$, using them to derive the support needed for intermediate artifacts such as task structures, spatial representations, and predictive rollouts. The primary differences among these methods lie in the intermediate artifacts through which that support is organized. We therefore classify cognitive planning into five paradigms (B1–B5) according to the primary intermediate artifact each paradigm produces. B1 produces executable semantic specifications. B2 organizes intent into logical task structures. B3 grounds intent in embodied context through intermediate task representations. B4 outputs spatial representations for direct execution. B5 imagines future dynamics for action evaluation. Among these paradigms, B1, B2, and B5 place greater emphasis on state transitions in $\mathcal{D}$, because their intermediate artifacts are organized around task progression and future evolution, whereas B3 and B4 place greater emphasis on sensory observations, because their intermediate artifacts remain grounded in the current embodied scene. Meanwhile, the intermediate artifacts associated with B1–B5 collectively suggest a broader trend toward forms that are increasingly grounded and closer to execution. Table 6 provides an overview of these paradigms.

4.1.1 Semantic Logic and Code Generation. B1 is closest to the semantic interface of MLLMs, outputting executable specifications such as code, reward functions, or skill rankings. Early work framed this as probabilistic alignment between language and affordance. SayCan [3] grounds abstract plans by scoring primitive skills against both semantic relevance and execution feasibility, but remains limited to atomic actions without compositional logic. Program synthesis goes further.

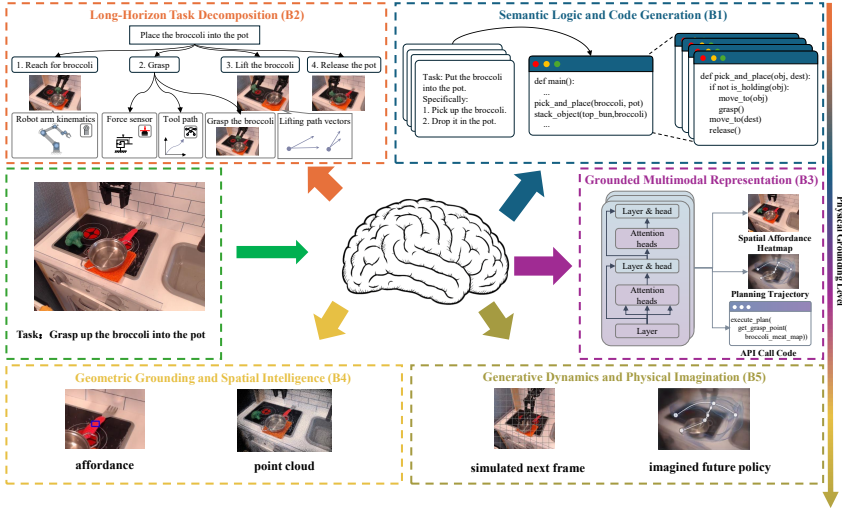


Fig. 5. Cognitive planning paradigms for embodied manipulation. The figure summarizes five representative planning modes: semantic logic and code generation, long-horizon task decomposition, grounded multimodal reasoning, geometric grounding and spatial intelligence, and generative dynamics and physical imagination.

CaP [102] generates executable Python from natural language, and ProgPrompt [155] embeds precondition assertions directly into the prompt structure to enforce logical constraints. A parallel line generates objective functions and leaves motor optimization to downstream solvers; Language to Reward [199] synthesizes programmatic reward functions that decouple semantic task specification from continuous trajectory optimization. Text2Touch [41] extends this paradigm to tactile dexterous manipulation, using LLM to design reward functions over tactile and proprioceptive signals for real-world manipulation. For structured control flow, LLM-BT [222] generates behavior trees that can be dynamically updated under environmental changes, enabling adaptive execution and recovery. LLM-as-BT [5] generates modular behavior trees that support reactive switching and error recovery. Demo2Code [171] extends the input modality by translating visual demonstrations into symbolic code, grounding program synthesis in physical observation alongside language. B1 primarily formulate task structure in the form of executable semantic specifications. For this reason, they mainly require $\mathcal{K}_2$ as their primary knowledge support.

4.1.2 Long-Horizon Task Decomposition. B2 converts intent into logical task structures, decomposing long-horizon goals into ordered subtask sequences or formal plans. LLM+P [107] addresses this by translating natural language into PDDL and delegating reasoning to classical solvers that guarantee feasibility. L3M+P [1] further enables the solver to learn domain transition models from experience, reducing reliance on pre-defined rules. Classical solvers, however, face combinatorial explosions in large environments. SayPlan [137] mitigates this by anchoring search in 3D scene graphs, hierarchically pruning irrelevant regions before planning. Inner Monologue [67] extends this paradigm to replanning with feedback, while Statler [196] maintains a task-relevant world state throughout execution. Since they depend on the organization of long-horizon task structure and subtask relations, they primarily require $\mathcal{K}_2$ as well. B1 and B2 paradigms share the limitation that their representations are insufficiently grounded in real physical execution.

4.1.3 Grounded Multimodal Representation. B3 grounds high-level task intent into intermediate task formulations that can be mapped onto executable skill APIs. PaLM-E [37] injects visual

Table 6. **Taxonomy of Cognitive Planning Paradigms.** **B1** = Semantic Logic & Code Generation; **B2** = Long-Horizon Task Decomposition; **B3** = Grounded Multimodal Reasoning; **B4** = Geometric Grounding & Spatial Intelligence; **B5** = Generative Dynamics & Physical Imagination. Output Abstraction denotes the generated artifact; Execution Gap specifies what additional computation converts the output into motor commands; Data Tier indicates the dataset level (Section 3.1) that primarily supports each paradigm.

ID	Support	Sub-direction	Output Abstraction	Execution Gap	Representative Methods
B1	$\mathcal{K}_2$	Probabilistic Alignment	Skill ranking	Skill API	SayCan [3]
		Program Synthesis	Executable code	Runtime interp.	CaP [102], Demo2Code [171]
		Reward Synthesis	Reward functions	RL optimizer	Language to Reward [199]
		Structured Control Flow	BT / XML	Reactive exec.	LLM-BT [222], LLM-as-BT [5]
B2	$\mathcal{K}_2$	Formal Solvers	Symbolic task plan	Classical planner	LLM+P [107], L3M+P [1]
		Semantic Search	Sub-task sequence	Motion planner	SayPlan [137]
		State-Aware Planning	Re-planned task state	Skill API	Huang et al. [67], Statler [196]
B3	$\mathcal{K}_2/\mathcal{K}_3$	Explicit Grounding	Text / action	Skill API	PaLM-E [37], RoboBrain [75]
		Implicit Grounding	CoT trace	Skill API	Mu et al. [124], ECoT [202]
B4	$\mathcal{K}_1/\mathcal{K}_2$	Physical Constraint	Contact / 6-DoF pose	Motion planner	ManipLLM [99], A3VLM [66]
		2D Affordance	Heatmap / mask	Grasp planner	CLIPort [153], PASG [228]
		3D Neural Fields	Neural field / 3D repr.	Motion planner	F3RM [149], D3Fields [173]
		Constraint Reasoning	Value map / keypts	Trajectory opt.	VoxPoser [68], ReKep [69]
B5	$\mathcal{K}_2/\mathcal{K}_3$	Future Video Prediction	Future frames	Inverse dynamics	UniPi [38], RoboDreamer [223]
		Latent World Simulation	Latent dynamics	Policy evaluation	UWM [226], V-JEPA2 [6]
		Physics-Constrained Prediction	Trajectory / latent dyn.	MPC / planner	Schulze et al. [143], [208], [127]

observations and continuous robot representation directly into the LLM embedding space, so that semantic reasoning is conditioned on the current embodied scene. EmbodiedGPT [124] extends this interface into the temporal domain by reasoning step by step with current observations in the loop, making task progression and intermediate subgoals more explicit. ECoT [202] externalizes richer intermediate traces, including task descriptions, object grounding, and plans every step, before action prediction. RoboBrain [75] pushes this paradigm toward a more integrated manipulation-oriented interface by jointly modeling planning, affordance, trajectory, and longer-horizon visual context. The primary knowledge support of B3 shifts from $\mathcal{K}_2$, associated with task-conditioned scene understanding, toward $\mathcal{K}_3$, associated with more transferable multimodal structure.

4.1.4 Geometric Grounding and Spatial Intelligence. B4 outputs spatial representations, including contact regions, 6-DoF poses, affordance maps, and neural fields, that planners and controllers can consume directly. ManipLLM [99] predicts pixel-wise contact masks from language, and A3VLM [66] extends this grounding to articulated objects by inferring joint axes and motion types. For 2D affordance mapping, CLIPort [153] combines semantic grounding with spatially precise manipulation for language-conditioned robot control., while PASG [228] strengthens the mapping with functional keypoints. For 3D affordance mapping, F3RM [149] injects open-vocabulary semantics into neural feature fields, while D3Fields [173] extends this line with dynamic 3D descriptor fields that encode semantic features and instance masks and support goal specification from 2D images. StructDiffusion [113] similarly maps high-level language goals and partial scene observations to physically valid target object arrangements for downstream execution. VoxPoser [68] and ReKep [69] push this line further by turning language or task descriptions into explicit 3D value maps and spatio-temporal relational keypoint constraints for downstream optimization. Across these methods, the common objective is to expose enough spatial structure for manipulation to query and optimize at execution time. B4 methods primarily represent the current geometric state of interaction, and therefore mainly require $\mathcal{K}_1$. When extended toward high-level constraint

reasoning, they increasingly require $\mathcal{K}_2$. Therefore, B4 representations cannot adequately capture state changes during contact, dynamic interaction, or long-horizon consequences.

4.1.5 Generative Dynamics and Physical Imagination. B5 comprises predictive planning artifacts, in which candidate actions are evaluated through explicit forecasts or imagined rollouts of future states, observations, or latent world dynamics. One route uses learned predictive models as the substrate for imagination. UniPi [38] reformulates policy as future video generation, UWM [226] unifies video and action diffusion within one transformer, and V-JEPA2 [6] shifts from pixel reconstruction to latent predictive reasoning. Purely learned prediction, however, drifts from physical consistency over long horizons. A second line constrains prediction with structural priors. Schulze et al. [143] inject analytical mechanics into learned dynamics, Zhang et al. [208] targets deformable interaction through particle-mesh structure, and LUMOS [127] uses language instructions to guide predictive rollouts in the learned world model. B5 methods primarily require $\mathcal{K}_2$ when they are used to organize state transitions through predictive rollout, they sometimes extend toward $\mathcal{K}_3$ by modeling broader latent predictive structure. However, B5 is limited by the reliability of predictive modeling itself. Therefore, errors accumulate in latent rollout, making it difficult to maintain stability over long horizons.

4.2 Policy Learning: The Embodied Cerebellum

Policy learning maps perceptual observations to motor commands, thus focusing more on the contact events and motion sequences. We organize these methods into three categories by how the action space is represented. P1 policies output continuous action vectors via regression in a single forward pass. P2 policies quantize actions into tokens and generate them autoregressively via cross-entropy. P3 policies model the full action distribution through multi-step denoising or flow matching. Within these three categories, sub-directions further distinguish how policy generation is factorized in practice, how it handles distribution shift, and how it generalizes across embodiments. Table 7 summarizes overview of this kind of paradigms, and Figure 6 contrasts these paradigms.

4.2.1 Explicit Continuous Policies. P1 policies map observations directly to continuous action vectors. The basic design separates a pre-trained visual encoder from a lightweight action head. BC-Z [72] uses this pattern for multi-task zero-shot generalization, MVP [136] shows that masked visual pre-training yields representations robust enough to reduce the policy to a simple MLP, and PointPolicy [51] extends explicit regression to 3D point clouds. Single-step prediction, however, ignores trajectory-level context. Decision Transformer [23] instead reframes control as autoregressive sequence modeling with a regression objective, while ACT-style methods [43, 217] combine action chunking with CVAE-based modeling to predict short-horizon action sequences and reduce jitter in bimanual and mobile manipulation. BAKU [50] combines CNN-based feature extraction with a transformer policy for multi-task settings. To handle distribution shift, MimicPlay [167] maps human play data to guide low-level visuomotor control trained on a small number of teleoperated demonstrations, Iterative Residual Policy [28] predicts residual corrections to recover from execution drift, and Motion Tracks [138] learns a unified motion representation for cross-embodiment adaptation. Within P1, reactive local control is grounded in local interaction and therefore primarily depends on $\mathcal{K}_1$. Task-conditioned, chunked policies, by contrast, rely more on task-level execution structure and accordingly require $\mathcal{K}_2$ more strongly. Despite their simplicity and efficiency, explicit continuous policies are poorly matched to multimodal action distributions, as a single deterministic prediction cannot capture multiple valid behaviors.

4.2.2 Discrete Action Policies. P2 policies discretize continuous actions into tokens and model them autoregressively, enabling multimodal action generation beyond unimodal regression. Early

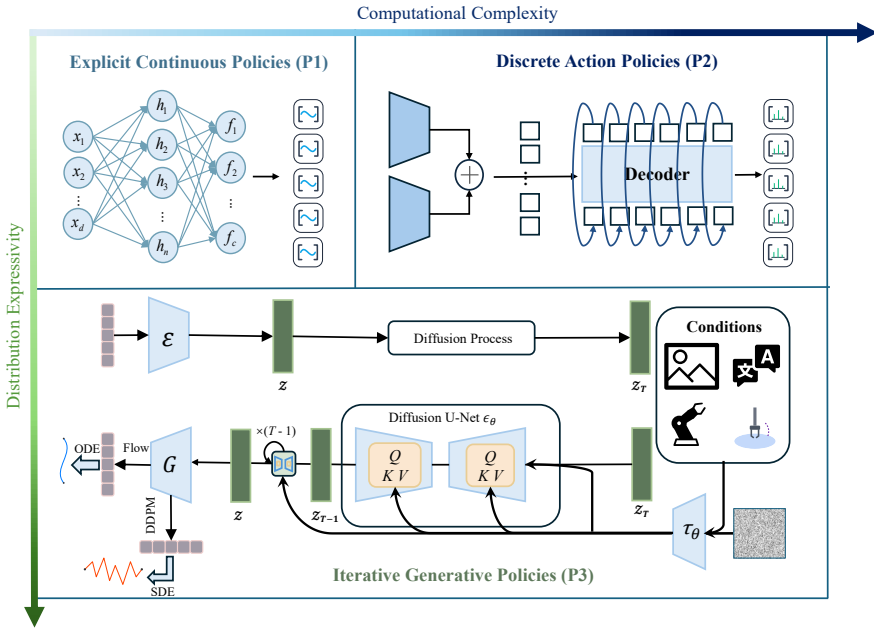


Fig. 6. Comparison of policy learning paradigms for robotic manipulation. The figure contrasts explicit continuous policies, discrete-action policies, and iterative generative policies in terms of distributional expressivity and computational complexity.

approaches employ uniform discretization. RT-1 [15] maps continuous dimensions to 256 bins, and PerAct [154] extends this to 3D by classifying actions over voxelized grids. However, fixed discretization imposes a rigid token space that is often poorly matched to continuous action distributions, motivating learned discretization methods that adapt the token space to the data. Learned discretization addresses this limitation. BeT [145] derives action modes through k-means clustering, VQ-BeT [91] learns a compact discrete action vocabulary with VQ-VAE, and CARP [46] further improves action prediction through multi-scale autoregressive modeling. Once actions are reliably tokenized, they can be generated within the same autoregressive framework used for text. VIMA [77] interleaves visual objects with text instructions for multimodal prompt conditioning, GR-1 [178] unifies vision, language, and action in a single stream, Seer [162] attributes much of the gain to the discrete formulation itself, and ARP [213] shows that chunk-level autoregressive generation generalizes across diverse robots. ICIL [42] further treats imitation as sequence completion, adapting to unseen tasks by conditioning on demonstration prompts without gradient updates. Within P2, task-scale discrete action generation remains organized around task-level action structure and therefore primarily requires $\mathcal{K}_2$. Multimodal and generalist token policies increasingly require $\mathcal{K}_3$ when they must organize transferable action structure across heterogeneous embodiments. The central challenge in these methods lies in aligning discrete action tokens with the structured, continuous nature of robot actions, while preserving action fidelity and efficient autoregressive generation.

4.2.3 Iterative Generative Policies. P3 policies treat action prediction as conditional generative modeling over continuous action trajectories, using iterative refinement to approximate the full action distribution. Early diffusion-based formulations first appeared in planning and structured

Table 7. **Taxonomy of Policy Learning Paradigms for Embodied Manipulation.** **P1** = Explicit Continuous (1-step forward pass, MSE/L1 regression); **P2** = Discrete Action (T -step autoregressive, cross-entropy); **P3** = Iterative Generative (K -step denoising/flow). Support indicates the dataset level (Section 3.1) that primarily supports each paradigm ($\mathcal{K}_1$ = Skill, $\mathcal{K}_2$ = Task, $\mathcal{K}_3$ = Foundation). Abbrev.: Percept. = Perception; Regr. = Regression; Resid. = Residual; Discr. = Discretization; Enhanc. = Enhancement; Consist. = Consistency; Diff. = Diffusion; refine. = refinement; Cont. = Continuous.

ID	Support	Sub-direction	Action Space	Scale	Input	Modeling	Methods
P1	$\mathcal{K}_1/\mathcal{K}_2$	Percept.-Action Regr.	Cont.	<100M	2D / 3D	Direct regression	BC-Z [72], [51]
		Temporal Chunking	Cont. chunk	100M-1B	2D / multi-view	Chunked action	ACT [217], BAKU [50]
		Recovery & Transfer	Cont. / resid.	100M-1B	2D / video	Goal-conditioned	MimicPlay [167], [138]
P2	$\mathcal{K}_2/\mathcal{K}_3$	Uniform & Spatial Discr.	Fixed bins	1B-55B	2D / voxel	Discr. categorical	RT-1 [15], Peract[154]
		Learned Quantization	Codebook	100M-1B	2D	Latent discrete	BeT [145], [91], [46]
		Prompted LM Tokens	LM tokens	1B-10B+	2D / multi-modal	Prompt-conditioned	VIMA [77], [178], [42]
		Autoregr. Generalists	LM tokens	1B-10B+	2D / multi-modal	Autoregr. sequence	Seer [162], ARP [213]
P3	$\mathcal{K}_1/\mathcal{K}_3$	Diffusion Foundations	Denoised cont.	<500M	2D	Diffusion denoising	Diffuser [74], [30]
		3D Geometric Enhanc.	Denoised cont.	<500M	3D / point cloud	Geometry-aware	DP3 [203], iDP3 [204]
		Flow & Acceleration	Flow cont.	<500M	2D / 3D	Flow / consistency	Prasad et al. [134], [188]
		Generalist Scaling	Denoised cont.	1B-8B	2D / multi-view	Scaled generative	Octo [130], [112], [139]

generation: Diffuser [74] casts long-horizon trajectory generation as probabilistic inpainting. Diffusion Policy [30] subsequently brings this paradigm into mainstream visuomotor control by learning the score function of the action distribution, thereby avoiding the mode-averaging limitations of direct regression. In 3D embodied manipulation, DP3 [203] further conditions diffusion on point-cloud observations for spatial grounding, while iDP3 [204] extends this formulation with improved 3D representations for more robust deployment. Despite these advances, iterative denoising still imposes substantial inference latency. Flow matching addresses this by learning velocity fields that transport noise to data along straighter trajectories with fewer integration steps. Consistency Policy [134] distills a teacher diffusion model into few-step generation, Streaming Flow Policy [76] streams action trajectories to the robot during generation, and RecFlow [188] accelerates inference by rectifying the learned flow itself. Among generalist robot policies, Octo [130] trains a transformer-based diffusion policy on the Open X-Embodiment dataset for cross-embodiment manipulation, RDT-1B [112] scales to the 1B-parameter regime on larger multi-robot data, and FLOWER [139] improves sample efficiency through flow-matching. Within P3, local generative control is grounded in continuous interaction and therefore primarily requires $\mathcal{K}_1$. Larger-scale generalist generative policies, however, require $\mathcal{K}_3$ to the extent that they are used to capture transferable action regularities across embodiments, rather than merely benefiting from large-scale pretraining. Overall, P3 policies offer the strongest distributional modeling capacity, but their practicality depends on making iterative generative control efficient enough for deployment while preserving action quality, control stability, and manageable compute cost.

4.3 Integration Models

To learn from the full structured record of $\mathcal{D}$, embodied manipulation systems often integrate cognitive planning and policy learning rather than relying on either path alone. RT-2 [229] provided an early demonstration that semantic understanding, reasoning, and action generation can be unified in this way, and later work has evolved into five recurrent integration patterns, which can be grouped into three coupling types. Type I emphasizes end-to-end unification within a shared backbone, as discussed in Section 4.3.1. Type II covers hierarchical designs that separate slower reasoning from faster execution, as discussed in Section 4.3.2. Type III maintains a shared

Table 8. **Integration patterns across data regimes.** The table summarizes recurrent integration patterns for embodied manipulation in terms of architecture type, supporting data regime, coupling principle, shared state, and typical pairings of cognitive planning and policy learning paradigms. Support indicates the data regimes that support each pattern. Paradigm codes follow Tables 6–7.

Pattern	Type	Data	Coupling Principle	Shared State	Typical B	Typical P
Explicit Hierarchy	II	$\mathcal{K}_2$	Explicitly decoupled	Language / code / task trace	B1 / B2	P1 / P2
Unified End-to-End	I	$\mathcal{K}_3$	Shared backbone	Minimal / implicit	B3	P2 / P3
Implicit Hierarchy	II	$\mathcal{K}_2/\mathcal{K}_3$	Implicitly decoupled	Latent / token / soft split	B3 / B2	P2 / P3
Spatial Grounding	III	$\mathcal{K}_1/\mathcal{K}_3$	Shared explicit geometric	Maintained spatial state	B4	P1 / P2 / P3
Predictive WM	III	$\mathcal{K}_2/\mathcal{K}_3$	Shared explicit predictive	Maintained predictive state	B5	P3

world representation that mediates between planning and policy, as discussed in Section 4.3.3. Table 8 summarizes these recurrent patterns in terms of architectural type, supporting data regime, coupling principle, shared state, and typical pairings of cognitive planning and policy learning paradigms, and Table 9 further lists representative integrated paradigms.

4.3.1 Type I: Unified End-to-End Foundation Models. Type I models treat actions as another modality tokenized alongside text and image patches, absorbing semantic understanding, reasoning, and action generation into a single backbone. Early paradigms represented actions as discrete tokens. RT-2 [229] co-fine-tuned a VLM to output discrete action tokens, and OpenVLA [88] released a 7B model trained on 970k Open X-Embodiment trajectories. X-VLA [219] continues this line with embodiment-specific soft prompts inside a shared transformer, showing how a unified backbone can extend across multiple robot platforms without introducing an explicit hierarchical split. Continuous representations followed. π_0 [14] introduced flow matching for high-frequency dexterous control, and RDT-2 [111] extends this to bimanual manipulation with zero-shot cross-embodiment deployment. RoboMamba [110] replaces self-attention with State Space Models for linear-time processing, and auxiliary reconstructive objectives encode 3D spatial structure to compensate for weak explicit geometry [157].

Although Type I paradigms demonstrate strong benefits from unifying perception, semantic reasoning, and action generation within a single architecture, this unification also comes at a cost. By forcing cognitive planning and motor control into the same representational and computational loop, Type I couples processes that operate at very different temporal and physical scales. The central challenge of this type is whether a single unified architecture can effectively support both high-level reasoning and real-time control.

4.3.2 Type II: Hierarchical & Modular Reasoning Models. Type II models decouple slow cognitive planning from fast policy execution. One branch uses explicit hierarchy. RT-H [13] introduced language motions as an intermediate layer between high-level instructions and low-level actions, CoT-VLA [216] and ThinkAct [65] expose intermediate reasoning steps, RoBridge [209] places a symbolic bridge between a high-level planner and an embodied agent, and HAMSTER [101] predicts a coarse 2D path as an intermediate representation to guide a low-level policy for precise 3D manipulation. Another branch uses implicit hierarchy, where the practical split is clear but the interface remains latent or only lightly structured. FiS-VLA [22] couples a slower semantic branch with a real-time execution branch, Hi Robot [151] adopts a hierarchical paradigm design for open-ended instruction following and user feedback during execution, and GO-1 [17] scales the same design to industrial bimanual control.

Table 9. **Representative Methods of Integration Models for Robotic Manipulation. Brain (B):** B1: Semantic Logic, B2: L-H Decomp, B3: Grounded MM Reas., B4: Geo. Grounding, B5: Generative Dynamics. **Policy (P):** P1: Expl. Continuous, P2: Discrete Action, P3: Iter. Generative. **Mode:** Dirc. = Direct Control; Reas. = Reasoned Control; Hier. = Hierarchical Control; Genr. = Generative Control. **Hardware:** Div. = Diverse multi-robot; Prop. = Proprietary. **Data Form:** Real Traj. = real robot trajectories; Web Vid. = internet video; Syn. = synthetic.

Model	B & P	Backbone & Params	Mode	Hardware	Primary Data Scale	Eval: Sim & Real
Type I: Unified End-to-End Foundation Models						
RT-2 [229]	B1, P2	PaLI-X (55B)	Dirc.	Google Robot	130K Real Traj.	Real: 700 manipulation tasks
OpenVLA [88]	B3, P2	Llama-2 + SigLIP (7B)	Dirc.	Franka+WidowX	970K Real Traj. (OXE)	Real: 29 manipulation tasks
$\pi 0$ [14]	B4, P3	PaliGemma (3B)	Genr.	Div. (6 types)	~10K Real Traj. (OXE)	Real: 7 dexterous tasks
RoboMamba [110]	B4, P2	Mamba SSM (7B)	Dirc.	Franka	OXE Real Traj.	Sim: CALVIN Real: manipulation tasks
UniVLA [18]	B5, P2	Emu3 (8.5B)	Dirc.	Div.	OXE Real Traj.	Sim: LIBERO (95.5%) Real: 10 multi-robot tasks
ReconVLA [157]	B3, P2	LLaVA-7B (Qwen2-7B)	Low	AgileX PiPer	2M samples (100K+ traj., BridgeV2/LIBERO)	Sim: CALVIN (ABCD→D) Real: 5 manipulation tasks
X-VLA [219]	B3, P3	Transformer (0.9B)	Genr.	Div.	290K episodes	Sim: LIBERO 93%, Simpler-WidowX Real: 3 robots
Type II: Hierarchical & Modular Reasoning Models						
RT-H [13]	B2, P2	PaLI-X (7B+)	Hier.	Google Robot	Kitchen Real Traj.	Real: 8 manipulation tasks
CoT-VLA [216]	B2, P2	VILA-U (7B)	Reas.	Div.	OXE Real Traj. + EPIC	Real: 10 long-horizon tasks
ThinkAct [65]	B1, P2	Qwen2.5-VL (7B)	Hier.	Franka	OXE + RoboVQA Real Traj.	Sim: CALVIN Real: 10 long-horizon tasks
Fis-VLA [22]	B2, P1	Llama-2 (7B)	Hier.	Franka	OXE Real Traj.	Real: 6 reactive tasks
HAMSTER [101]	B5, P1	VILA-1.5 (13B)	Hier.	Franka	Off-domain Web Vid.	Sim: Colosseum Real: 74 tasks / 222 evals
Hi Robot [151]	B2, P3	PaliGemma (3B)	Hier.	Div.	Syn. data	Real: 10 manipulation tasks
RoBridge [209]	B2, P1	VLM Planner	Hier.	Gen3	Syn. + Real Traj.	Real: 12 manipulation tasks
Type III: Spatial Grounding & Predictive World Models						
3D-VLA [218]	B4, P3	Flan-T5-XL (3B)	Genr.	Franka	2M samples OXE (3D)	Sim: RL Bench Real: 10 3D tasks
VideoVLA [150]	B5, P3	CogVideoX (5B)	Genr.	Div.	OXE + 5K samples	Sim: SIMPLER Real: 6 unseen tasks
SpatialVLA [135]	B4, P2	PaLiGemma 2 (4B)	Dirc.	Franka+WidowX	OXE + RH20T Real Traj.	Sim: RL Bench Real: 10+ spatial tasks
TraceVLA [220]	B4, P3	Llama-2 + SigLIP (7B)	Dirc.	WidowX	OXE + 150K Real Traj.	Sim: SimplerEnv (137 configs) Real: 4 WidowX tasks
DreamVLA [211]	B5, P3	Diffusion WM (7B)	Genr.	Franka	CALVIN Real Traj.	Sim: CALVIN Real: 10 manipulation tasks
BridgeVLA [97]	B4, P2	PaliGemma (3B)	Dirc.	Franka	Bridge / OXE Real Traj.	Real: 10 3D tasks
DreamZero [195]	B5, P3	Wan2.1 (14B)	Genr.	Div.	Web Vid. + Robot Traj.	Real: 2× generalization

Although explicit and implicit Type II models offer advantages, their drawbacks are also evident. Explicit hierarchies risk forcing task-relevant information into fixed intermediate representations, which can lead to the loss of execution-sensitive information. Implicit hierarchies, by contrast, can hide coordination across levels in opaque internal states, making them difficult to supervise or correct. What they share is the bottleneck introduced by decomposition: once slow planning is separated from fast control, the interface between cognition and policy becomes the critical constraint on information flow, error correction, and coordination across levels.

4.3.3 Type III: Spatial Grounding & Predictive World Models. Type III maintains a shared world representation that planning and policy jointly consume. This world state takes two forms. Spatial

Table 10. Deployment bottlenecks and representative techniques across the 3-Stage pipeline.
S1 = Offline Pre-deployment; **S2** = Online Physical Alignment; **S3** = Inference-Time Execution.

Stage	Deployment Bottleneck	Technique Category	Representative Methods
S1	Offline RL at Scale	Autoregressive Q-fn, offline TD backups	Q-Transformer [21], TRANSIC [78]
	Cross-Embodiment	R2S2R latent action spaces	X-Sim [34]
	Sim-to-Real & Compression	Delta action compensation, DCT tokenization	ASAP [53], FAST [133]
S2	RL Fine-Tuning	Direct GRPO, world-model-guided adaptation	SimpleVLA-RL [96], DiWA [20]
	Contact-Rich Adapt.	Consistency policies, visuo-tactile refinement	ConRFT [27], VT-Refine [64]
	Forgetting & Residual	Off-policy residual RL	PLD [184], OpenVLA-OFT [87]
S3	Quantization	Action-centric mixed-precision, 1-bit PTQ	QVLA [187]
	Decoding Acceleration	Parallel fixed-point iterations	TinyVLA [177], PD-VLA [156]
	Robustness & Safety	Contrastive decoding, motion planning	Policy CD [180], Neural MP [33]
	Failure Detection & Recovery	Flow-based OOD detection, runtime updates	FAIL-Detect [185], Reflective TTP [60]

grounding paradigms build a persistent geometric representation: 3D-VLA [218] embeds 3D perception into the LLM for spatial reasoning, SpatialVLA [135] grounds actions in metric 3D structure via Ego3D Position Encoding, TraceVLA [220] conditions generation on visual trajectory traces, and BridgeVLA [97] aligns spatial and predictive representations to ensure kinematic feasibility. Predictive world models instead learn forward dynamics from data: VideoVLA [150] jointly predicts action sequences and future visual outcomes using a pre-trained video generator, DreamVLA [211] integrates a diffusion-based world model into the manipulation pipeline for planning via imagined trajectories, and DreamZero [195] learns a joint predictive model over future visual states and actions.

Although Type III models exhibit strong potential, they rely heavily on a reliable shared world representation. If this representation is inaccurate or stale, the entire system can be misled. While correction is possible, it often incurs substantial computational and engineering cost. In addition, Type III models typically demand large-scale data and resources to train, and still face significant bottlenecks in real-world deployment.

4.4 Deployment Pipeline

The embodiment foundations define the feasible physical space that constrains how manipulation capability can be realized. In practice, many methods introduce additional processing during deployment before learned models can be executed reliably in the real world. These processing steps are increasingly being organized into a deployment pipeline. Before deployment, models are refined and prepared for physical rollout through offline adaptation, transfer, and compression. During deployment, they are further aligned to the target hardware and interaction conditions. At runtime, they are executed under real-world constraints with support for efficient, robust, and safe control. Table 10 summarizes the main bottlenecks and representative techniques at each stage.

4.4.1 Stage 1: Offline Pre-deployment. Before physical deployment, the main challenges are transferring policies from simulation to reality, generalizing across embodiments, and making large models practical for real-world execution. Q-Transformer [21] shows that autoregressive Q-functions can scale offline actor-critic learning to transformer policies through temporal difference backups on static datasets. X-Sim [34] uses a real-to-sim-to-real pipeline to learn cross-embodiment manipulation from human videos by treating object motion as a dense and transferable signal. ASAP [53] mitigates dynamics mismatch between simulation and reality by learning a residual action compensation model that aligns simulated dynamics with real-world execution. TRANSIC [78] adopts a complementary human-in-the-loop strategy in which operators provide online corrections during

real-world execution, from which a residual policy is learned and combined with the original simulation policy for autonomous deployment. As model size grows, pre-deployment compression also becomes necessary: FAST [133] improves deployment efficiency by replacing naive action discretization with a compressed action representation based on the discrete cosine transform.

4.4.2 Stage 2: Online Physical Alignment. Once deployed on physical hardware, the primary bottleneck shifts from transfer preparation to adaptation under real contact, friction, sensing noise, and OOD objects. SimpleVLA-RL [96] shows that large manipulation models can be fine-tuned directly through reinforcement learning, FLRe [63] addresses the tension between generalist priors and specialist performance through adaptive large-scale RL fine-tuning, while DiWA [20] adapts diffusion policies through an offline world model, reducing the amount of physical interaction needed for post-training improvement. ConRFT [27] combines offline BC and Q-learning with online consistency policies and human intervention checkpoints to support safe adaptation in contact-rich settings. Human-in-the-loop real-world reinforcement learning provides another effective route for this stage: HIL-SERL [115] combines demonstrations, learned reward modeling, and online human corrections to learn precise and dexterous manipulation skills directly on physical robots. VT-Refine [64] further demonstrates that visuo-tactile policies, when paired with tactile digital twins and refinement in simulation, can improve high-precision dual-arm coordination. Other alignment strategies include residual adaptation via off-policy RL [184]. The adaptation choices made here directly affect Stage 3: aggressive full-parameter fine-tuning can improve task performance but complicate compression and serving, whereas residual or modular updates preserve more of the original policy structure at the cost of additional execution-time overhead.

4.4.3 Stage 3: Inference-Time Execution. During real-world deployment, the key challenge is to sustain reliable closed-loop execution on edge hardware. QVLA [187] addresses action-sensitive quantization through channel-wise bit allocation, assigning higher precision to channels that are more sensitive in the action space. TinyVLA [177] shows that compact paradigm design can improve both inference speed and data efficiency, even eliminating the need for a separate large-scale pretraining stage, while PD-VLA [156] accelerates chunked action generation by replacing sequential autoregressive decoding with parallel decoding iterations. Runtime robustness is equally critical. Policy Contrastive Decoding [180] mitigates sensitivity to irrelevant visual features during action generation, FAIL-Detect [185] reframes runtime failure detection as sequential OOD detection without requiring failure data, and Neural MP [33] improves execution reliability by distilling classical motion planning into a fast reactive neural planner that generalizes across cluttered real-world settings. Reflective TTP [60] supports online recovery through reflection and updates during test, re-evaluating actions with hindsight to adapt the policy during deployment.

Despite progress across all three stages, current methods still face several limitations. Pre-deployment approaches often remain sensitive to simulator fidelity, offline data coverage, and the scope of embodiment variation, which limits the robustness of transfer to open-ended real-world settings. Online physical alignment methods improve adaptation under real dynamics, but they often require expensive robot interaction, careful reward or intervention design, and, in many cases, substantial human involvement that may hinder scalability. At inference time, compression, quantization, and decoding acceleration improve deployability, yet their impact on fine-grained action fidelity and contact-rich control are insufficiently understood.

5 System Evaluation of Robotic Arms

Previous sections have examined how robotic manipulation systems are shaped by embodied foundations, data, learning, and deployment. This chapter reviews current efforts to evaluate the resulting systems from four corresponding perspectives. Section 5.1 evaluates whether cognitive

Table 11. Evaluation metrics and representative works for embodied manipulation.

Dimension	Metric Category	Representative Metrics	Representative Works
Execution Quality	Task Completion	Stage Success, Task Progression	RoboEval [174], [144]
	Trajectory Smoothness	SPARC, LDLJ, Path Length, Jerk	SPARC [10], LDLJ [4], [174]
	Dual-Arm Coordination	Velocity Divergence, Height Discrepancy	RoboEval [174]
Data Efficiency & Quality	Info-theoretic	Mutual Information, Demo Contribution	DemInf [55]
	Gradient-based	Influence Scores, Hessian-based Valuation	CUPID [2], QoQ [90]
	Sample Efficiency	AULC	General IL benchmarks
Generalization & Transfer	Human Preference	Pairwise Comparison, VLM-based Eval	RoboArena [7], [225]
	OOD Generalization	ΔSR_{OOD} , RMSD	REALM [144]
	Sim-to-Real & World Models	MMRV, VLM-as-Judge, ΔSR	SIMPLER [100], [48], [147]
Deployment Efficiency	Energy Decomposition	P_{elec} , $P_{friction}$, P_{copper} , P_{mech}	ECDP [57]
	Normalized Efficiency	$\eta_E(\mathbf{q})$, $f_U/f_C/f_R$, CoT	Vidussi et al. [164], [58]
	Multi-objective Optimization	Execution Time, Jerk, Energy	Rezali et al. [140]

planning (B) and policy learning (P) can organize the knowledge support $\mathcal{K}$ induced by $\mathcal{D}$ into successful embodied task execution. Section 5.2 evaluates whether the structured record $\mathcal{D}$ provides data that is sufficiently efficient and informative to induce useful knowledge support $\mathcal{K}$. Section 5.3 evaluates whether the knowledge support $\mathcal{K}$ learned from $\mathcal{D}$, and organized through B and P , remains robust beyond the training distribution under varying embodiment constraints. Section 5.4 evaluates whether the embodied capability produced by B and P from $\mathcal{K}$ can be executed efficiently on physical embodiments within the deployment pipeline. Table 11 summarizes representative metrics and protocols.

5.1 Task Execution Quality Evaluation

Task execution is a widely adopted evaluation criterion because it provides a direct measure of model success on the task. However, binary success rates are easy to interpret, but they cannot distinguish early failure from almost successful and unstable execution, especially in long-horizon or bimanual settings. Finer-grained metrics can be grouped into three categories. Task progression metrics are more closely aligned with $\mathcal{K}_2$ -level task organization, trajectory quality metrics probe $\mathcal{K}_1$ -level execution quality, and bimanual coordination metrics reflect execution coupling. For task completion, RoboEval [174] decomposes manipulation into sequential stages and measures progress as the fraction completed, REALM [144] further evaluates robustness under perturbations using the Root Mean Squared Deviation of task progression relative to the default setting:

$$\text{RMSD}(p) = \sqrt{\frac{1}{MT} \sum_{m=1}^M \sum_{t=1}^T \left(r_{m,t}^p - r_{m,t}^{\text{default}} \right)^2}, \quad (1)$$

where $r_{m,t}$ denote the task progression for trial p under perturbed and default conditions. For trajectory quality, smoothness metrics such as SPARC [10] and LDLJ [4] quantify jerk and motion regularity, while Dynamic Time Warping measures alignment with expert demonstrations under temporal misalignment. For bimanual manipulation, RoboEval also measures coordination through velocity divergence and height discrepancy between end effectors.

5.2 Data Efficiency and Quality Evaluation

Data efficiency and quality concern whether the structured record $\mathcal{D}$ is sufficiently informative and efficient to induce useful $\mathcal{K}$. Existing evaluation efforts in this direction are typically conducted along three dimensions. One dimension is whether a dataset provides broad coverage enough for

knowledge induction. DemInf [55] estimates the contribution of each demonstration to the mutual information between states and actions, using this quantity to capture both state diversity and action predictability. Another dimension is whether demonstrations remain useful for downstream learning. CUPID [2] estimates how individual demonstrations affect downstream policy performance, providing a way to rank training data by utility. QoQ [90] instead evaluates samples by their contribution to validation improvement, helping identify low-quality or detrimental demonstrations. Data efficiency is commonly summarized by the Area Under the Learning Curve:

$$\text{AULC} = \frac{1}{N_{\max}} \int_0^{N_{\max}} \text{SR}(n) \, dn \approx \frac{1}{N_{\max}} \sum_k \text{SR}(n_k) \Delta n_k \quad (2)$$

where $\text{SR}(n)$ is the success rate after training on n demonstrations. AULC is more informative than reporting final success after a fixed data budget, since it captures how quickly useful capability emerges.

5.3 Generalization and Transfer Evaluation

Generalization and transfer evaluation concern whether the knowledge support $\mathcal{K}$ learned from the structured record $\mathcal{D}$, and organized through B and P , remains valid under distribution shift and varying embodiment conditions. Existing efforts in this direction can be grouped into three categories: human judgment, robustness under distribution shift, and proxy validity. Human judgment evaluates aspects of manipulation quality that are not fully reflected by automated success metrics. RoboArena [7] uses pairwise human comparison to infer global policy rankings across distributed real-world evaluations, while AutoEval [225] reduces human evaluation overhead by automating success detection and scene resets, thereby minimizing the need for manual trial-by-trial assessment. Robustness under distribution shift measures whether performance remains stable outside the training distribution. REALM [144] evaluates generalization under visual, semantic, and behavioral perturbations, using changes in task progression and the Root Mean Squared Deviation of task progression to quantify degradation under shift, and further reporting binary success rate and time to completion at the task level. Proxy validity concerns whether scalable evaluation environments preserve the same conclusions as real-world execution. SIMPLER [100] measures this through Mean Maximum Rank Violation,

$$\text{MMRV}(R, R_S) = \frac{1}{N} \sum_{i=1}^N \max_{1 \leq j \leq N} (|R_i - R_j| \mathbf{1}[(R_{S,i} < R_{S,j}) \neq (R_i < R_j)]) \quad (3)$$

where R_i and $R_{S,i}$ denote the real-world and simulated performance of policy i , respectively. WorldArena [147] extends this idea to learned world models by asking whether they can serve as reliable evaluators in addition to simulators. Together, these methods define a hierarchy of increasingly scalable but increasingly approximate proxies, from the real world to simulators to learned world models.

5.4 Deployment Efficiency Readiness Evaluation

Deployment efficiency readiness concerns whether the embodied capability produced by B and P from $\mathcal{K}$ can be executed efficiently on physical embodiments under real deployment constraints. Relevant costs include latency, compute, memory, hardware load, and energy consumption. Existing evaluation efforts in this direction can be grouped into three categories: energy decomposition, normalized efficiency comparison, and multi-objective trade-off analysis. Existing energy evaluations usually begin with simplified power models or their proxies before moving to more structured decompositions. Another structured view is provided by ECDP [57], which decomposes total power

into electronics, friction, copper loss, and useful mechanical work:

$$P_{\text{total}}(t) = P_{\text{elec}} + P_{\text{friction}}(t) + P_{\text{copper}}(t) + P_{\text{mech}}(t) \quad (4)$$

This decomposition is useful because it attributes where energy is spent rather than reporting only a single aggregate number. It also shows that control motors can dominate total consumption, which implies that trajectory optimization alone may have limited impact on overall energy efficiency. For comparison across robots, recent work introduces normalized efficiency metrics [58, 164] and measures such as Cost of Transport, while Rezali et al. [140] formulate trajectory planning as a multi-objective optimization problem over several metrics. Even so, deployment efficiency remains only partially quantified in current embodied manipulation benchmarks, since latency, runtime reliability, compute footprint, and hardware load are still rarely reported in comparable ways.

Overall, existing evaluation efforts do not suffice to close the validation loop across the overall pipeline. A comprehensive characterization of how $\mathcal{D}$, $\mathcal{K}$, embodiment, and the organization of B and P determine behavior under real deployment conditions is lacking.

6 Open Challenges and Future Directions

Building on the foregoing organizational framework, we examine the open challenges of embodied manipulation through the constraint chain developed above. We show how limitations originating in embodied foundations accumulate in data support, intensify in architectural design, and ultimately make evaluation an unstable source of feedback for upstream stages. On this basis, we then examine safety and deployment of deployed embodied models, where additional problems emerge under coupled constraints beyond those discussed above. Finally, we conclude the section with a brief discussion of several emerging directions.

6.1 Embodiment Constraints

Current embodied systems remain confined to narrow observation and action spaces induced by their embodiment. This limitation leaves richer embodied modalities only weakly supported. Moreover, substantial differences in mechanical parameters, control interfaces, and data formats mean that current manipulation systems remain strongly tied to platform-specific priors. The open challenge, therefore, is to expand embodied support while establishing transferable representations, interfaces, and data abstractions that generalize across embodiments. UniDex [206] reflects one attempt to organize data, action spaces, and learning more coherently across embodiments, while IAIL [26] suggests that transfer may need to be grounded at the level of intention rather than trajectory correspondence. However, neither direction yet resolves the deeper non-equivalence across embodiments, and the core challenge remains open.

6.2 Data Support

To support a richer and more reliable $\mathcal{K}$, $\mathcal{D}$ must more comprehensively, consistently, and reproducibly capture motion sequences, state transitions, sensory observations, and contact events. Teleoperation can provide best data coverage but such data collection remains slow, expensive, and difficult to scale. By contrast, UMI improves portability and collection efficiency at the cost of weaker support for sensory observations and contact events. More scalable forms of autonomous data collection can alleviate this bottleneck, but they are reliable only for interactions that can be repeated, reset, and executed safely at scale. Therefore, autonomous collection datasets are unable to provide sufficient coverage for state transitions and contact events in $\mathcal{D}$. Simulation and synthetic datasets are often easier and cheaper to generate, but the $\mathcal{K}$ they induce is often misaligned with real-world manipulation. Physics engines still face substantial difficulty in accurately modeling contact, friction, compliance, and deformable dynamics. These limitations are especially

evident in important real-world settings such as failures, recoveries, contact-rich transitions, and cross-embodiment adaptation. As a result, whether simulated data can provide reliable support for downstream manipulation learning remains unresolved.

6.3 Architectural Design

Paradigms for robotic arm manipulation are shaped by the structural biases imposed by embodied foundations, while the capabilities they provide are still derived from incomplete and often unreliable $\mathcal{K}$. Structural biases induced by embodied foundations prevent the field from converging on a single dominant architecture, so that architectural design becomes increasingly heterogeneous and tightly coupled to embodied constraints. Incomplete $\mathcal{K}$ further amplifies this heterogeneity by preventing any single paradigm from simultaneously optimizing abstraction, executability, efficiency, and transfer. Even a broad family such as the policy paradigms summarized in Table 7 therefore continues to split into subdirections, each reflecting a different engineering trade-off. Under these conditions, recent architectural variation increasingly takes the form of different ways of organizing and coupling existing components within a shared design space. This growing heterogeneity is reflected not only in the proliferation of broad labels such as VA, VLA, WM, VAM, and WAM, but also in the increasing instability of how planning, prediction, and action are connected across systems. As these functions are combined in increasingly heterogeneous ways, they become less separable and more tightly coupled across stages, because the information and constraints needed downstream cannot be reliably preserved through a unified interface. Such tightly coupled designs are therefore difficult to reuse and extend reliably in real-world settings.

6.4 Evaluation

Existing metrics have become more fine-grained for execution quality, data efficiency, and transfer validity, but they are still concentrated around a relatively narrow set of task formats and embodiments. Existing evaluation frameworks remain centered on narrow and idealized task settings, with metrics that are still too limited to adequately capture robustness, generalization, and real-world deployment performance. Therefore, existing evaluation provides only a partial view of the capabilities that matter for real-world deployment. Existing benchmark environments are often built around preferred task formats, observation settings, and evaluation criteria. As a result, they tend to privilege capabilities that are easy to standardize and compare, while offering only limited visibility into the robustness, transferability, and long-horizon interaction skills required in real deployment. Once a small set of such benchmark suites becomes dominant, these implicit preferences begin to shape the evaluation ecosystem more broadly. Research often adapts to their task distributions, observation formats, and success criteria, which can in turn steer model development toward benchmark compatibility over the broader capabilities required for embodied manipulation. Real-world evaluation poses a separate reproducibility challenge. Results collected in different laboratories are rarely directly comparable, even when task descriptions appear similar, because embodiment, calibration, sensor placement, scene layout, intervention policy, and environmental conditions often differ substantially. Without stronger shared physical protocols and standardized real-world test procedures, validation on physical robots is still fragmented and often difficult to reproduce across settings. Recent efforts such as ManipArena [158] make the problem explicit: existing evaluation remains split between simulation benchmarks that are controllable but incomplete proxies for deployment, and real-world testing that is more realistic but still fragmented and weakly reproducible across platforms.

6.5 Trustworthiness Under Coupled Constraints

Even when embodiment, data, and learning are tightly integrated in deployed embodied systems, trustworthiness remains a distinct and unresolved challenge. In learned manipulation systems, this coupling introduces unique risks that are not captured by task performance alone. BadVLA, for example, shows that strong performance on benign inputs does not guarantee secure deployment, as robotic manipulation systems can still exhibit control deviations in the presence of a backdoor trigger [224]. More broadly, this suggests that embodied systems should be evaluated not only by whether they complete tasks, but also by whether internally hidden failures that can propagate into physical behavior can propagate into unsafe or untrustworthy physical behavior. At the same time, policy learning paradigms provide limited visibility into how perception is mapped into action. When failures occur, it is often difficult to determine whether they arise from observation encoding, reasoning, action representation, or mismatches with downstream controllers [108]. Hierarchical and chain-of-thought paradigms offer more visible intermediate structure, but these intermediates do not necessarily reveal the true basis of the final decision. Trustworthiness is further complicated in shared human environments, where successful deployment depends not only on task completion, but also on whether people can anticipate system behavior, interpret its intent, and intervene safely when necessary. Current embodied paradigms still provide limited support for such kind of calibrated human-robot collaboration, leaving unresolved questions around responsibility, liability, and acceptable deployment boundaries.

6.6 Emerging Directions

Recent work suggests that the field is beginning to move beyond isolated improvements toward broader cross-stage directions. On the embodiment and platform side, platforms such as UniDex [206] highlight a shift toward embodiment-aware frameworks that unify data generation, cross-embodiment policy learning, and evaluation for dexterous manipulation. IAIL [26], by contrast, aligns behavior in terms of intention, allowing robots with different morphologies to learn from one another without requiring trajectory correspondence. At the same time, data pipelines are becoming more reusable. Always-on egocentric capture [190] explores how everyday human activity can be turned into a scalable data stream without dedicated recording sessions, while Tether [103] shows how single demonstrations can be adapted across novel scenes without relying on a physics engine. Together, these developments suggest a shift away from narrowly scoped, platform-specific experience toward representations and data pipelines that generalize across hardware, environments, and tasks. What becomes transferable in these systems is no longer limited to trajectories or task-specific demonstrations, but increasingly includes intentions, action abstractions, and intermediate representations that can be reused across embodiments, environments, and deployment stages. The relationship between learning and deployment is also changing. Embodied-R1 [200] shows that progress in reasoning and action modeling can matter as much as scale, while VLANeXt [?] shows that principled architectural design within a unified embodied architecture can also lead to substantial performance gains. PF-DAG [92] shows that hybrid policy design is a promising direction, decoupling coarse mode selection from fine-grained continuous action generation. WoVR [79] and ReinFlow [210] move beyond static pretraining through post-training with world models and online reinforcement learning. These studies suggest that embodied agents can continue to adapt after deployment. As embodied systems move closer to real-world use, evaluation rigor and deployment trustworthiness are increasingly important. RoboChallenge [189] advances standardized real-world evaluation for embodied policies. SafeVLA [205] focuses on safety alignment under deployment conditions. Häon et al. [52] emphasize transparency and steerability through mechanistic interpretability.

7 Conclusion

This review has argued that robotic arm manipulation is better understood as a process of capability formation under embodiment constraints than as a collection of expanding model paradigms. From this perspective, some seemingly methodological differences can be read as different responses to the same coupled conditions across embodiment, data, learning, deployment, and evaluation. On this basis, the field should be organized not only by model labels, but also by how different methods are situated within the broader process of capability formation.

At the current stage, further progress depends less on adding new loosely specified categories than on strengthening shared support: clearer methodological scope, more reproducible evaluation protocols, and scalable close-loop ecosystems that can be reused across the community. Strengthening these shared foundations will be important for improving comparability across studies and enabling more cumulative progress across the community. We hope this review can provide a useful reference for future research on robotic arm manipulation.

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